

Ananya Prabhakar

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Objective

A motivated Information Science Engineering student with solid knowledge of software development and object-oriented programming. Skilled in Python, Java, SQL, and web technologies, with hands-on experience through academic projects. Passionate about building efficient, scalable applications and learning new technologies.

Education

- Jawaharlal Nehru New College of Engineering, Shimoga** 2022-2026
BE in Information Science Engineering
CGPA: 8.38 / 10
- Government P U College, Thirthahalli** 2019-2022
Senior Secondary Education
92.33 %
- Government Girls High School, Thirthahalli** 2017-2019
Metriculation
87.84 %

Skills

- Programming languages: C, Java, Python, JavaScript, React.js
- Web Technologies: HTML, CSS, JavaScript
- Data bases: MySQL, MongoDB
- Operating Systems: Windows, Linux, Ubuntu
- Data Structures & Algorithms
- Softwares & Tools: VS Code, Eclipse, Spyder, Git, GitHub, Oracle
- Visualization: Tableau, Power BI

Projects

- Academic Projects**
- ECG Signal Analysis and Interpretation**
 - Developed an ECG prediction system by preprocessing ECG images using denoising, segmentation, normalization, and resizing techniques.
 - Implemented and evaluated multiple models, with the CNN model achieving the most reliable prediction performance based on accuracy, precision, recall, and confusion matrix metrics.
 - Deployed the trained CNN model using Python, TensorFlow/Keras, OpenCV, and Flask, enabling real-time ECG prediction through a web interface.
- Heart Disease Prediction using Machine Learning**
 - Developed a heart disease prediction system trained on clinical datasets containing features such as age, sex, chest pain type, blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, old peak, and ST slope.
 - Implemented and evaluated Decision Tree, Random Forest, and SVM models with structured data preprocessing and feature engineering, using accuracy, precision, recall, F1-score, and AUC-ROC metrics.
 - Designed a Heart Disease Prediction Interface where user-entered clinical values generate real-time predictions indicating Heart Disease or No Heart Disease using Python, Scikit-learn, TensorFlow, Pandas, and NumPy.
- Bank Loan Approval Prediction using Logistic Regression**
 - Developed a machine learning-based prediction system using Logistic Regression to determine whether a customer will accept a personal loan (1) or not (0) based on demographic and financial attributes such as age,

income, education, family size, and credit card spending.

- Performed data preprocessing, feature scaling, and model training using Python and Scikit-learn, and evaluated model performance through accuracy, precision, recall, F1-score, and confusion matrix analysis.
- Achieved 95.5% prediction accuracy, with visual insights generated using correlation heatmaps and performance plots, demonstrating the model's effectiveness for real-world banking decision support.
- **Forward Search Attack on RSA-Based Online Voting System**
 - Designed and implemented an RSA-based online voting system to study security vulnerabilities arising from deterministic RSA encryption without padding, focusing on voter privacy risks.
 - Simulated a Forward Search Attack by encrypting all possible voting options using the public key and exploiting server Yes/No validation responses to infer the voter's choice without breaking RSA.
 - Demonstrated effective mitigation using RSA-OAEP padding, introducing randomness into encryption to prevent ciphertext matching and protect vote confidentiality.

Training

- 30-Days Classroom Training Program by Fresher Profiles conducted at JNNCE Campus, Shivamogga. Covered core topics including Python programming, Java programming, aptitude (quantitative, logical reasoning, verbal), and soft skills development. Improved problem-solving ability, coding fundamentals, logical thinking, and professional communication skills for entry-level IT roles.

Achievements And Awards

- Chess Competency (District Level): Competed in district-level chess tournaments, demonstrating strong problem-solving ability, logical reasoning, strategic thinking, and sustained focus under competitive conditions.
- Rural IT Quiz Participant: Participated in a Rural IT Quiz, strengthening analytical skills, logical thinking, and understanding of core IT concepts and digital literacy.
Competitive Aptitude: Demonstrated consistent problem-solving skills, analytical thinking, and technical aptitude through participation in academic and competitive technical events.
- Publications:
"ECG Signal Analysis and Interpretation Using Machine Learning", International Journal of All Research Education and Scientific Methods (IJARESM), Vol. 13, Issue 12, 2025.

Interests

- Software Development & Programming (Python, Java)
- Database Management (SQL, NoSQL)
- Data Analysis & Visualization
- Web Development

Hobbies

- Playing Chess
- Drawing
- Playing badminton
- Solving puzzles

Languages

- Kannada(Native)
- English